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CSC1001 final 2024

Computer Science (香港中文大學（深圳））

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Desk Number _____

Student Number _____

Student Name _____

School of Science and Engineering

FINAL EXAMINATION

2024

CSC1001 Introduction to Computer Science

Examination Duration:

Reading Time:

This examination has ____ questions.

Exam Conditions:

This is a FORMAL Examination

This is a RESTRICTED OPEN BOOK Exam. Maximum of one (1) sheet of handwritten notes double sided are permitted

Materials Permitted In The Exam Venue:

Maximum of one (1) sheet of handwritten notes double sided are permitted. **NO OTHER MATERIALS PERMITTED**

Any calculators without the functionalities of programming and file storage are permitted.

Materials To Be Supplied To Students:

1 × 7 Page Answer Booklet

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Question 1. (36%)

Please give the output for each program as below.

	Program	Output

1)	`x = 1+2**3/4*(5+2)`
	`s = "3*5//6+7"`
	`print(eval(s+str(x)))`
		717
		3 points

2)	`day = 12`
	`s = "0123456789"`
	`days = [0,1,2,3,4,5,6,7,8,9]`
	`flag = (str(day) in s) and (day in days)`
	`print(flag)`
		False
		3 points

3)	`x = 10`
	`def reset_x(x):`
	`  x = 20`
	`  return x`
	`reset_x(x)`
	
	`print(x)`
		10
		3 points

4)	`n1, n2 = 10, -10`
	`for i in [1,2,3]:`
	`  print(n1-i, n2+i, sep = ',')`
	`  n1, n2 = n2+2, n1-2`
		9，-9
		-10,10 **1 point per line**
		7，-7

5)	`n = 6`
	`def f(n = 3)`
	`  while n<6:`
	`  print("Part 1: n is", n)`
	`  n+=3`
	`  return n`
	`  print("Part 2: n is", n)`
	`  n+=3`
	`  print("Part 3: n is", n)`
	`print(f())`
		Part 1： n is 3 （1 point）
		6 （2 points）

6)	`A = [[1,2,0],[0,1,2],[1,2,3],[1,2,0]]`
	`x = 0`
	`for i in range(1,4,2):`
	`  for j in A[i]:`
	`  x = x + A[i][j]`
	`print(x)`
		6
		3 points

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7)	`def foo(m, n):`
	`  if m==0:`
	`  return n+1`
	`  elif m>0 and n==0:`
	`  return foo(m-1,1)`
	`  elif m>0 and n>0:`
	`  return foo(m-1, foo(m, n-1))`
	
	`print(foo(foo(1,0), 1))`
		5
		
		3 points

8)	`d = {'Alex':{'CSC1001':'A+',`
	`  'CSC1002':2}, 'Bob':['CSC1001',`
	`  96, 'CSC1002', 90]}`
	
	`print(d['Alex'][d['Bob'][0]])`
	`print(d['Bob'][d['Alex']['CSC1002']][:3])`
		A+ 1 point
		
		CSC 2 points

9)	`Dict = {'aaa':1, 'aab':10, 'cde':15, 'zoo':200}`
	`Seq = Dict.items()`
	`L1 = list()`
	`L2 = []`
	
	`for k,v in Seq:`
	`  L1.append((k,v))`
	`  L2.append((v,k))`
	
	`L1.sort()`
	`print(L1[2])`
	`L1 = sorted(L2)`
	`print(L1[2])`
		（'cde', 15）1 point
		
		(15, 'cde') 2 points

10)	`class A:`
	`  def __init__(self, R = 5):`
	`  self.r = R`
	`  self.R = 0`
	
	`  def setR(self, r = 5):`
	`  self.R = r`
	
	`a = A()`
	`print(a.R, a.r)`
	`a.setR(10)`
	`print(a.r, a.R)`
		0 5 1 point
		
		5 10 2 points

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11)	<pre>class A:
	  def __getInfo(self):
	  return "A"
	  def printInfo(self):
	  print(self.__getInfo())
	class B(A):
	  def __getInfo(self):
	  return "B"
	
	A().printInfo()
	B().printInfo()</pre>
		A 1 point
		
		A 2 points

12)	<pre>class A:
	  def __init__(self):
	  self.str = "CSC1001"
	  def output(self)
	  print("the string in A is:", self.str)
	
	class B(A):
	  def __init__(self):
	  super().__init__()
	  self.str = self.str + "CSC1002"
	  def output(self)
	  print("the string in B is:", self.str)
	
	class C(B):
	  def __init__(self):
	  super().__init__()
	  self.str = self.str + "CSC1003"
	
	c = C()
	c.output()</pre>
		the string in B is: CSC1001CSC1002CSC1003
		
		3 points

Multiple correct implementations might exist for programming questions below; rubrics and examples provided just for reference.

Question 2. (3*4%=12%)

Please answer the following sub-questions.

1) Please use recursion to implement the function power(x, n).

Correct recursive implementation (3 points)

Note: Either linear recursion OR divide-and-conquer recursion is acceptable, but must have corresponding correct time complexity in (2)

Correct base case (1 point)

2) Please give the complexity analysis of your implementation in 1).

Correct answer (O(logn) or O(n)) based on student's implementation in (1) (4 points)

3) Please write a program to print all elements in a singly linked list.

Correctly initialize pointer from list head (1 point)

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Loop with correct termination condition, e.g., While p!=None: (2 points)

Update pointer to next node, e.g., p = p.next (1 point)

Example:

```
def print_list(L):
  p = L.head
  while p != None:
    print(p.data)
    p = p.next
```

Question 3. (15%)

Please fill in the following blocks (A-D) to complete the programs.

1) **BinarySearch (2*3%)**

```
# the function takes a sorted list "alist" and an item as input,
# if the item can be found in the list, return true, or return false
def binarySearch(alist, item):
    if len(alist) == 0:
        return False
    else:
        midpoint = len(alist)//2

    if alist[midpoint]==item:
        return True
    else:
        if item<alist[midpoint]:
        ##############################
            # Block A
        ##############################
        else:
        ##############################
            # Block B
        ##############################
```

Block A: return binarySearch(alist[: midpoint], item) (3 points)

Block B: return binarySearch(alist[midpoint:], item) (3 points)

2) **MergeSort (5%+4%)**

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```python
# Merges two sub-lists of A[].
# First sub-list is A[l..m]
# Second sub-list is A[m+1..r]
def merge(A, left, mid, right):
    n1 = mid - left + 1
    n2 = right- mid
    # Copy data to temp Aays L[] and R[]
    for i in range(0 , n1):
        L.append(A[left + i])

    for j in range(0 , n2):
        R.append(A[mid + 1 + j])

    # Merge the temp lists back into A[left..right]
    i,j,k = 0,0,1   # Initial index

    while i < n1 and j < n2 :
        ############################
          # Block C
        ############################

    # Copy the remaining elements of L[]
    while i < n1:
        A[k] = L[i]
        i,k = i+1, k+1
    # Copy the remaining elements of R[]
    while j < n2:
        A[k] = R[j]
        j,k = j+1, k+1
    return A
```

```python
# left is for left index and right is right index of the
# sub-list of A to be sorted
def mergeSort(A,left,right):
    if left < right:

        mid = (left+(right-1))/2

        # Sort first and second halves and merge them together
        ############################
          # Block D
        ############################


# the sample code to run mergeSort for sorting a list
a = [12, 11, 13, 5, 6, 7]
n = len(a)
mergeSort(a,0,n-1)
print ("\n\nSorted list is")
for i in range(n):
    print ("%d" %a[i]),
```


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Block C:

```
if L[i] < R[j]:
    A[left+k-1] = L[i]
    i, k = i+1, k+1
else:
    A[left+k-1] = R[j]
    j, k = j+1, k+1
```

if else -- 1 point

remaining 4 lines 1 point per line

Block D:

```
mergeSort(A, left, mid)
mergeSort(A, mid+1, right)
merge(A, left, mid, right)
```

mergeSort(A, left, mid) 1 point

mergeSort(A, mid+1, right) 1 point

merge(A, left, mid, right) 2 points

Question 4. (10%)

Given a list of integers, return **indices** of the two numbers such that they add up to a specific target. You may assume that each input would have *exactly* one solution, and you may not use the *same* element twice.

Correct nested for loops (4 points)

- Partial credit: single for loop (1 point)

Check if two elements sum to target (2 points)

Correctly return indices (2 points)

Avoid using same element twice (2 points)

Question 5. (12%)

Given a singly linked list L, where L.head is a reference of its first element and L.tail refer to its last element. For each element p, p.value stores an integer number and p.next refers to the reference of its next element. Concerning a number v, please write a program to reorganize the list to make all elements smaller than v are in the front of the remaining elements. One example is shown below, where v=30 reorganized the list into two parts. Note that, it is not allowed to create a new list, please just reorganize the original list.

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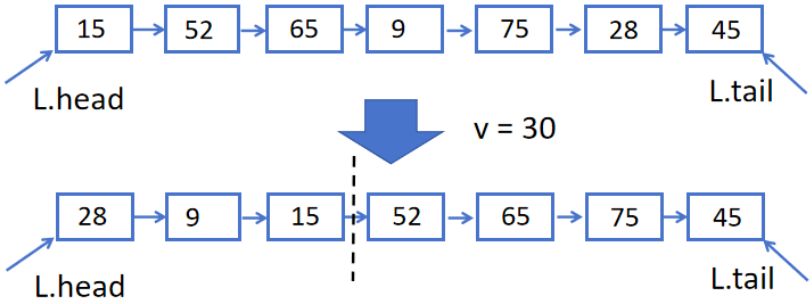


```
p = L.head

q = p.next              # 2 points

while q != None:            # 2 points

   if q.value < v:          # 2 points

      p = p.next           # 1 point

      q.next = L.head

      L.head = q           # 2 points

      q = p.next           # 1 point

   else:

      p = p.next
      q = p.next           # 2 points
```

Question 6. (15%)

Please design a class named "Mat3Multipy". The class should provide functions to load matrices (3*3) from files, to do multiplication and to save the result to a new file. Note: the matrices in the file are in 3 lines, and 3 numbers each line.

Correct Class and method definition format (4 points)

__init__() method (2 points)

load method (3 points):

- File opening (1 point)

- String to number conversion (1 point)

- Form matrix (1 point)

Multiply method (4 points):

- Triple nested loops (3 points)

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- Correct formula (1 point)

Save method (2 points)

Example (Implementation may vary; any logically correct solution is acceptable):

```
class Mat3Multiply:                    # Class definition (4 points)
    def __init__(self):                # __init__ method (2 points)
        self.matrix = []

    def load(self, filename):          # load method (3 points total)
        with open(filename, 'r') as f:     # File opening (1 point)
            self.matrix = []
            for line in f:
                row = [int(x) for x in line.split()]  # String to number (1 point)
                self.matrix.append(row)     # Form matrix (1 point)

    def multiply(self, other):         # Multiply method (4 points total)
        result = [[0, 0, 0], [0, 0, 0], [0, 0, 0]]
        for i in range(3):
            for j in range(3):
                for k in range(3):         # Triple nested loops (3 points)
                    result[i][j] += self.matrix[i][k] * other.matrix[k][j]  # Formula (1 point)
        return result

    def save(self, result, filename):      # Save method (2 points)
        with open(filename, 'w') as f:
            for row in result:
                f.write(' '.join(map(str, row)) + '\n')
```


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END OF EXAMINATION

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